

Bitcoin and Blockchain Technologies

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Origins of Bitcoin: The Pre-History

Why the World Needed Bitcoin

The Double-Spending Problem

Digital cash dilemma: a digital coin is just a file — Alice can copy it and pay both Bob and Charlie simultaneously

Traditional fix: rely on a trusted bank to verify each payment (centralised, expensive, censorship-prone)

Nakamoto's insight: replace the trusted third party with a decentralised ledger updated by cryptographic consensus

Failed Early Attempts

DigiCash (1989): blind-signature e-cash; required a central bank-like server; bankrupted 1998

E-gold (1996–2009): gold-backed digital tokens; seized by US regulators

Hashcash (1997): proof-of-work to fight email spam — later the core of Bitcoin mining

Intellectual Lineage

b-money (1998): Wei Dai proposed anonymous distributed e-cash; Satoshi cited it directly

Bit Gold (1998): Nick Szabo designed a decentralised gold analogue with proof-of-work and ownership chains

RPoW (2004): Hal Finney's Reusable Proofs of Work — closest precursor to Bitcoin's reward mechanism

Key Lesson

Bitcoin synthesised 15 years of cypherpunk research into one working system — solving the double-spending problem without a trusted centre.

History of Bitcoin: The White Paper and Genesis

October 2008: Satoshi Nakamoto publishes “Bitcoin: A Peer-to-Peer Electronic Cash System” on the cryptography mailing list — nine pages that changed finance

January 3, 2009: Genesis block mined; embedded headline: “*The Times 03/Jan/2009 Chancellor on brink of second bailout for banks*” — a political message in code

January 12, 2009: First transaction ever: Satoshi sends 10 BTC to Hal Finney

May 22, 2010: Laszlo Hanyecz buys 2 pizzas for 10,000 BTC (“Bitcoin Pizza Day”) — first real-world commerce

April 2011: Satoshi hands over project to Gavin Andresen and disappears with the words: “I’ve moved on to other things. It’s in good hands.”



The White Paper and Satoshi's Identity

Pseudonym: “Satoshi Nakamoto” — one person or a team, never confirmed; name means roughly “wise central origin” in Japanese

Released: October 2008, active until April 2011, then silent forever

Proposed suspects: Hal Finney, Nick Szabo, Dorian Nakamoto, Craig Wright — none proven

Bitcoin stash: early mining suggests Satoshi holds ~1 million BTC; none have ever moved (worth ~\$90B at 2024 prices)

Deliberate disappearance: removing a leader keeps Bitcoin leaderless and censorship-resistant — possibly a feature, not a bug

Bitcoin: A Peer-to-Peer Electronic Cash System

Satoshi Nakamoto
satoshin@gmx.com
www.bitcoin.org

Abstract. A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted third party is still required to prevent double-spending. We propose a solution to the double-spending problem using a peer-to-peer network. The network timestamps transactions by hashing them into an ongoing chain of hash-based proof-of-work, forming a record that cannot be changed without redoing the proof-of-work. The longest chain not only serves as proof of the sequence of events witnessed, but proof that it came from the largest pool of CPU power. As long as a majority of CPU power is controlled by nodes that are not cooperating to attack the network, they'll generate the longest chain and outpace attackers. The network itself requires minimal structure. Messages are broadcast on a best effort basis, and nodes can leave and rejoin the network at will, accepting the longest proof-of-work chain as proof of what happened while they were gone.

Notable Participants: The Founding Community

Hal Finney — Cypherpunk Pioneer

Built **Reusable Proofs of Work (RPoW)** in 2004, the direct precursor to Bitcoin mining

Received the first on-chain transaction (10 BTC) from Satoshi on January 12, 2009

Hands-on debugging of **Bitcoin v0.1**; died 2014 from ALS; body cryogenically preserved

Wei Dai and Adam Back

Wei Dai: author of b-money (1998); Satoshi wrote: *“I’m expanding on your ideas. . .”*

Adam Back: Hashcash inventor; directed Satoshi to b-money; now CEO of Blockstream

Gavin Andresen — Lead Maintainer

Took over as lead developer when Satoshi stepped back in 2011

Satoshi’s final email: “please don’t frame me as a mysterious figure — give credit to the broader team”

Laszlo Hanyecz — GPU Miner

Built the first GPU miner; Satoshi asked him to **slow down** so CPU users could still participate

Paid 10,000 BTC for two pizzas (May 2010) — worth ~\$900M at peak prices

Key Lesson

Bitcoin was not built by one genius alone. A small cypherpunk community, each with specific expertise, made the launch possible.

Satoshi's Emails: The Evidence

Satoshi Nakamoto [satoshi at vistomail.com](mailto:satoshi@vistomail.com)

Fri Oct 31 14:10:00 EDT 2008

- Previous message: [Fw: SHA-3 lounge](#)
- **Messages sorted by:** [\[date\]](#) [\[thread\]](#) [\[subject\]](#) [\[author\]](#)

I've been working on a new electronic cash system that's fully peer-to-peer, with no trusted third party.

The paper is available at:

<http://www.bitcoin.org/bitcoin.pdf>

The main properties:

Double-spending is prevented with a peer-to-peer network.
No mint or other trusted parties.

Satoshi's Emails: The Final Handoff

Satoshi Nakamoto <satoshin@gmx.com>
Mike Hearn <mike@plan99.net>

Sat, Apr 23, 2011 at 3:40 PM To:

I had a few other things on my mind (as always). One is, are you planning on rejoining the community at some point (eg for code reviews), or is your plan to permanently step back from the limelight?

I've moved on to other things. It's in good hands with Gavin and everyone.

I do hope your BitcoinJ continues to be developed into an alternative client. It gives Java devs something to work on, and it's easier with a simpler foundation that doesn't have to do everything. It'll get critical mass when impatient new users can get started using it while the other one is still downloading the block chain.

Bitcoin: Key Facts and Market Data

What Bitcoin Is

Decentralised digital currency: no central bank, no issuer, no owner

Fixed supply: capped at exactly **21 million BTC** — 19.7M already mined as of 2024; last coin around year 2140

Smallest unit: 1 **satoshi** = 0.00000001 BTC (one hundred-millionth); named after Nakamoto

Key innovation: solves double-spending without a trusted third party using **blockchain** and **proof-of-work**

Supply Schedule (The Halving)

Started at **50 BTC/block** (2009); halves every 210,000 blocks (≈ 4 years)

Market Context

All-time high: $\sim \$108,000$ (December 2024)

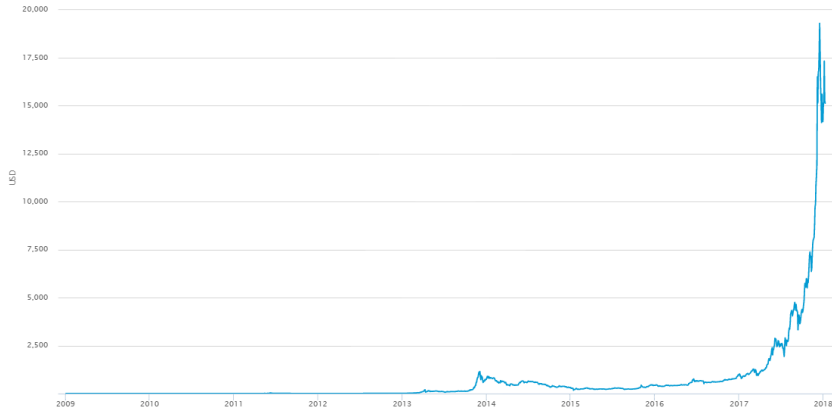
Market cap: $\sim \$2$ trillion at peak — larger than most national stock exchanges

Daily transactions: $\approx 300,000$ – $500,000$ per day; confirms $\approx$ every 10 minutes

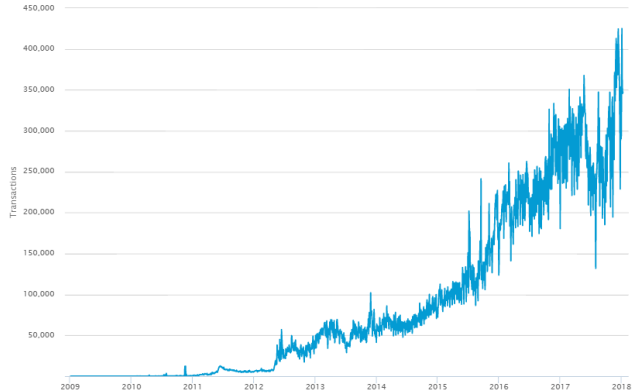
Key Lesson

Bitcoin's scarcity is *mathematically enforced* — no government can inflate it away. The supply schedule is hard-coded and has never been changed in 15 years of operation.

Bitcoin Price History



Bitcoin Transactions (Daily)



Cryptography and Blockchain Foundations

Desired Features of Digital Currencies

Security Properties

Unforgeability: one cannot create coins out of thin air or copy them like a file

Exclusive ownership: only the owner (holder of the private key) can spend funds

Pseudonymity: addresses reveal no personal identity; transactions are publicly visible but not trivially attributable

Double-spend prevention: the same coin cannot be spent twice; requires distributed consensus

Usability Properties

Divisibility: Bitcoin is divisible to 8 decimal places (satoshis)

Fungibility: every BTC is interchangeable with every other BTC

Speed: transactions broadcast in seconds; confirmed in ~10 minutes

Decentralisation: no single point of failure; thousands of full nodes worldwide

Key Lesson

All these properties are achieved through cryptography, not through trust in any institution. This is Bitcoin's fundamental engineering innovation.

The SHA-256 Hash Function: An Intuition

What a Hash Function Does

Takes **any input** (a word, a document, a block of transactions) and produces a **fixed-length fingerprint**

SHA-256: output is always exactly **256 bits** (64 hexadecimal characters) regardless of input size

One-way: given the hash, you cannot recover the input (computationally infeasible)

Collision-resistant: it is infeasible to find two different inputs with the same hash

Avalanche Effect: Tiny Change, Huge Difference

“Alice sent Bob 2.5 bitcoins?” → b711226...

“Alice sent Bob 2.5 bitcoins.” → c0b1b09...

“Alice sent Bob 2.51 bitcoins?” → 053e749...

Why This Matters for Bitcoin

Every block stores the **SHA-256 hash of the previous block** — changing any old transaction would change all hashes downstream

Mining is a **SHA-256 puzzle**: find a number (nonce) so the block hash starts with many zeros

Addresses are derived from **double SHA-256** of the public key

Key Lesson

SHA-256 is Bitcoin's workhorse. It appears in mining, in address generation, in the Merkle tree, and in the chain linking. Bitcoin uses it twice (SHA-256d) for extra security.

Cryptographic Hash Functions: A Survey

Known Hash Functions

MD5: 128-bit output; *broken* — collisions found since 2005; do not use for security

SHA-1: 160-bit output; *broken* — Google and CWI Amsterdam found a collision in 2017 (the “SHattered” attack) using 9.2 quintillion SHA-1 computations

SHA-256 (SHA-2 family): 256-bit output; *still secure*; used in Bitcoin, TLS, and most modern systems

SHA-3 (Keccak): published by NIST in 2015; Ethereum uses Keccak-256 (a variant)

Applications Beyond Bitcoin

Commitments: Alice hashes her answer and timestamps it; reveals the preimage later to prove priority

File integrity: software downloads include a SHA-256 checksum — compute it yourself to detect tampering

Password storage: websites store hashes, not passwords; a breach reveals only hashes, not plaintext (with proper salting)

Digital signatures: message is hashed first, then signed — allows signing of any-length message

Collision Resistance of SHA-256



Code: Computing SHA-256 in Python

```
import hashlib

def sha256(text: str) -> str:
    return hashlib.sha256(text.encode()).hexdigest()

# Avalanche effect: tiny change -> completely different hash
msg1 = "Alice sent Bob 2.5 bitcoins?"
msg2 = "Alice sent Bob 2.5 bitcoins." # just a period change
msg3 = "Alice sent Bob 2.51 bitcoins?"

print(sha256(msg1)) # b711226415fd343dc413cf730272...
print(sha256(msg2)) # c0b1b093d80e88c3ac836f0d25de...
print(sha256(msg3)) # 053e74934a04d0d0fe09a3f5d10e...

# Double SHA-256 (Bitcoin's standard -- used in block headers and addresses)
def sha256d(text: str) -> str:
    first = hashlib.sha256(text.encode()).digest()
    second = hashlib.sha256(first).hexdigest()
    return second

print("SHA256d:", sha256d("Bitcoin genesis block"))
```

Blockchain: A Tamper-Evident Linked List

Structure of a Blockchain

A **blockchain** is a linked list of **blocks**; each block contains:
Transaction data (the “payload”)

Hash of the previous block (the “chain link”)

A **nonce** and timestamp

The first block is the **genesis block** (no previous hash)

Why It Is Tamper-Evident

Change any byte in block 500 $\Rightarrow$ block 500's hash changes
 $\Rightarrow$ block 501's prev-hash is wrong $\Rightarrow$ all subsequent blocks invalid

To forge history: redo **proof-of-work for every subsequent block** — impossible for Bitcoin (would cost billions)

Bitcoin Block Contents

Block header: version, prev-hash, Merkle root, timestamp, difficulty target, nonce

Block body: all transactions in that block (typically 1,500–3,000)

Block size limit: 1 MB (legacy) up to 4 MB with SegWit; target ≈ 10 min/block

Block reward: 3.125 BTC (post-April 2024) plus all transaction fees

Key Lesson

The blockchain does not prevent theft — it prevents *undetectable alteration*. Every full node independently verifies the entire chain from genesis onward.

Merkle Trees: Efficient Transaction Verification

What Is a Merkle Tree?

A **binary tree of hashes**: leaf nodes are transaction hashes; parent nodes are hashes of their two children

The **Merkle root** (single top hash) summarises all transactions and is stored in the block header

Any change to any transaction changes the Merkle root — immediately detectable

Merkle Proofs: Verifying Without Full Data

SPV: a lightweight wallet verifies a transaction with $O(\log n)$ hashes — no need to download all transactions

With 4,000 transactions per block, only 12 hashes needed to prove inclusion

Why Merkle Trees Matter

Efficiency: mobile wallets verify payments without the full blockchain (≈ 600 GB as of 2024)

Integrity: changing any transaction requires recomputing the Merkle root and re-mining the block

Used everywhere: Git, IPFS, Ethereum state trie, and certificate transparency logs all use Merkle structures

Key Lesson

The Merkle tree lets Bitcoin be *efficient and secure simultaneously*. SPV wallets verify billions in payments using just kilobytes of data.

Code: Building a Merkle Tree in Python

```
import hashlib

def sha256d(data: str) -> str:
    h = hashlib.sha256(data.encode()).digest()
    return hashlib.sha256(h).hexdigest()

def merkle_root(txids: list) -> str:
    """Compute the Merkle root of a list of transaction IDs."""
    if len(txids) == 1:
        return txids[0]
    # If odd number, duplicate last element (Bitcoin protocol rule)
    if len(txids) % 2 == 1:
        txids.append(txids[-1])
    parents = []
    for i in range(0, len(txids), 2):
        combined = txids[i] + txids[i+1]
        parents.append(sha256d(combined))
    return merkle_root(parents)

# Example: 4 transactions in a block
transactions = ["tx_alice_to_bob", "tx_carol_to_dave",
               "tx_eve_to_frank", "tx_grace_to_heidi"]
txids = [sha256d(tx) for tx in transactions]
root = merkle_root(txids)
print("Merkle Root:", root[:20], "...")
# Change any one transaction -> root changes completely
```

Digital Signatures and Bitcoin Addresses

Digital Signatures: The Ownership Mechanism

How Digital Signatures Work

Key pair: every user generates a **private key** (secret, 256 bits) and a **public key** (shareable)

To **send** Bitcoin: sign with your private key — proves you authorise the spend

Anyone can **verify** using your public key — no need to reveal the private key

Without the private key, **signatures cannot be forged** — even with the public key and many past signatures

ECDSA in Bitcoin

Elliptic Curve DSA (ECDSA) on curve *secp256k1*

Private key: **256 bits**; public key: **512 bits**; signature: **512 bits**

Message is hashed (SHA-256) before signing — works for any message length

Critical Security Rule

Never reuse a nonce in ECDSA signing: in 2010, PlayStation 3 was hacked because Sony reused a nonce — private key recovered from two signatures

Random key generation is critical: entropy must be truly random; hardware wallets use dedicated random number generators

Private key theft = permanent, irreversible loss; Bitcoin transactions cannot be reversed

Key Lesson

A Bitcoin private key is literally your money. Anyone who has it can spend your coins. Unlike a bank, there is no password reset and no customer service.

Digital Signature: ECDSA Diagram

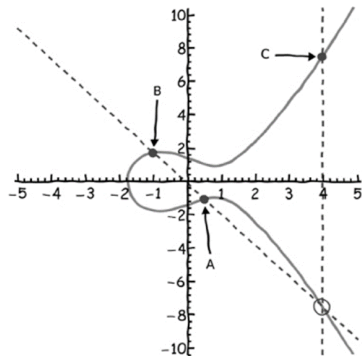
Private key (256 bits): secret scalar on the elliptic curve; generated from a random number

Public key (512 bits): a point on the secp256k1 curve; derived by multiplying the private key by the generator point G

Signature: pair of numbers (r, s) derived from the private key and message hash; verifiable with the public key

One-way: you can compute public key from private key, but not vice versa (discrete logarithm problem)

Source: Mastering Bitcoin, Antonopoulos



Bitcoin Addresses: From Key to Address

A **Bitcoin address** is not your public key — it is a **hashed, encoded version** of it

Steps: Private key → Public key (ECDSA) → SHA-256 → RIPEMD-160 (160 bits) → Base58Check encoding

Result: a 25–34 character string starting with 1 (P2PKH) or 3 (P2SH) or bc1 (SegWit)

Genesis block address:

1A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa (the “Satoshi address” with ~70 BTC never spent)

Can also be represented as a **QR code** for easy scanning at point of sale



Online RSA Key Generator

Key Size 512 bit

Generate New Keys

Generated in 10 ms

☐ Async

Private Key

```
-----BEGIN RSA PRIVATE KEY-----
MIIBOgIBAAJBAKKVNEGAmdYb6jzyCubUQI6MkiW9SnLRs4d5YRGznMm0j3Tlne/
8L
hcVoHjg8mFDyN/EiULY1rebweRpZzuG771cCAwEAAQJAYIEciSUEutzhnFY0N0N
HT0w1CG5Hyyi2JpLIIBK7nNogN7ETTtaZiP4DUnJP86bqTaNamwz07I021eeI
iQIhAM6jZxRMojU6x0YZXS$2tkmNoy8xhRR4BjUKDFAiz1AiEAyWuqYxxG++Ni
diO6koGK1h16192KdmK4gzTfz25cIQCPnRnSvzWITPZ23F1c0XznG+RySj8BS
4Gg/WpO3UAr2bQlqSqV8WdJuu+WnsrBFRrToVGgNOaeyOBmmDk96GJQbssECI
Eg6
RWI8KIWhV4AsQY8KURdzhkug1TFI88pOPuqYDhWI
-----END RSA PRIVATE KEY-----
```

Public Key

```
-----BEGIN PUBLIC KEY-----
MFwwDQYJKoZIhvcNAQEBBQADSwAwSAJBAAKKVNEGAmdYb6jzyCubUQI6Mki
W9SnLRs
4d5YRGznMm0j3Tlne/8LhcVoHjg8mFDyN/EiULY1rebweRpZzuG771cCAwEAAQ=
=
-----END PUBLIC KEY-----
```

RSA Encryption Test

Text to encrypt:

This is a test

Encrypt / Decrypt

Encrypted:

Xsr4YVJg4D3+5Jgwb/HSmnM7rj8RVL2KvRAzdA4UYbi9jW8I2mtYsIF
Hk+dlxX9m1Q4NttvSpO61KQj1fUZg==



Open Source JavaScript Client-Side Bitcoin Wallet Generator

69%	69%	69%	Brain Wallet
69%	69%	Wallet Details	

Generating Bitcoin Address...

MOVE your mouse around to add some extra randomness... 69%

OR type some random characters into this textbox

abb69f05fdd1f3a0c18ec7c57fa1189b6f8d36f5899add7351a26eba16d5e2cebdbb4ba
22444191f746efee12c806dc23ab52f2a252b766a65561128a12f8dbba2f8b12ba5d154
b9a8fe8ef363d8872ae928250c256465fc141b46ca59bfff2766919715fe3d0105ea90e5
6e5ab7fd97e543b7a75213de01d4a21e5e3e720f3d6c98d8d46eff89cb44fc6fdeec4c
c7d114bbfbb6e7005c33018b57f700baaa8907722340b85bedaf037a86f714d2441b8c7
26e880b49a087360737a8e042e7f843b0e313a1d663bde5345648c259e9cafa38d0621
a3ed5b8b740bc187b14b6d69395b59248d813bd9be2ce82d149ab69002336f60dc50eec
4d934ad1a287e0e



Donations: **1NINja1bUmhSoTXozBRBEiR8LeF9TGbZBN**
[GitHub Repository \(zip\)](#)

[Version History \(3.3.0\)](#)

527B 5C82 B1F0 B2DB 72A0
ECBF 8749 7B91 6397 4F5A
[\(PGP\)](#) [\(sig\)](#)

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Single Wallet

Paper Wallet

Bulk Wallet

Brain Wallet

Vanity Wallet

Split Wallet

Wallet Details

Generate New Address

Print

Bitcoin Address

Private Key



SHARE



SECRET

1Q85vY8tRpZQVV3JurXRzy1wdK6sRwdcpY

L2AqK1rKE4d4DDs2pj5oruYRrv7VBHG4HP3KqhpsTNCQsPLZpZYW

A **Bitcoin wallet** is as simple as a single pairing of a Bitcoin address with its corresponding Bitcoin private key. Such a wallet has been generated for you in your web browser and is displayed above.

To safeguard this wallet you must print or otherwise record the Bitcoin address and private key. It is important to make a backup copy of the private key and store it in a safe location. This site does not have knowledge of your private key. If you are familiar with PGP you can download this all-in-one HTML page and check that you have an authentic version from the author of this site by matching the SHA256 hash of this HTML with the SHA256 hash available in the signed version history document linked on the footer of this site. If you leave/refresh the site or press the "Generate New Address" button then a new private key will be generated and the previously displayed private key will not be retrievable. Your Bitcoin private key should be kept a secret. Whomever you share the private key with has access to spend all the bitcoins associated with that address. If you print your wallet then store it in a zip lock bag to keep it safe from water. Treat a paper wallet like cash.

Add funds to this wallet by instructing others to send bitcoins to your Bitcoin address.

Check your balance by going to blockchain.info or blockexplorer.com and entering your Bitcoin address.

Spend your bitcoins by going to blockchain.info and sweep the full balance of your private key into your account at their website. You can also spend your funds by downloading one of the popular bitcoin p2p clients and importing your private key to the p2p client wallet. Keep in mind when you import your single key to a bitcoin p2p client and spend funds your key will be bundled with other private keys in the p2p client wallet. When you perform a transaction your change will be sent to another bitcoin address within the p2p client wallet. You must then backup the p2p client wallet and keep it safe as your remaining bitcoins will be stored there. Satoshi advised that one should never delete a wallet.

Code: Generating a Bitcoin Address from a Private Key

```
import hashlib, struct

# Step 1: Define a private key (hex) -- NEVER use this in production!
private_key_hex = "20de0bec7e51b4cbdd4b6baeedef3de15491db94f83c3f883400b75dc950dcc3"

# Step 2: (In practice, use a library like 'bitcoin' or 'ecdsa')
# pip install bitcoin
from bitcoin import privkey_to_pubkey, pubkey_to_address

private_key = private_key_hex
public_key = privkey_to_pubkey(private_key)
address = pubkey_to_address(public_key)

print("Private Key:", private_key_hex[:20], "...")
print("Public Key :", public_key[:20], "...")
print("Address      :", address)
# Output: mi8Bx79yfdWD27uSrtfVc4PCAB2HifyoKR (testnet address)

# The address hides the public key -- quantum computers could break ECDSA
# but would still need to reverse RIPEMD-160 to get the address preimage
```

Bitcoin Wallets: Hot and Cold Storage

Hot Storage (Online)

Exchange wallets (Coinbase, Binance): easiest to use; custodial — the exchange holds your keys; vulnerable to exchange hacks (Mt. Gox: \$450M lost, 2014; FTX: \$8B, 2022)

Software wallets (Electrum, Bitcoin Core): you hold the keys; vulnerable to malware on the device

Mobile wallets: convenient; attack surface includes phone theft and SIM swapping

Cold Storage (Offline)

Hardware wallets (Ledger, Trezor): private keys never touch the internet; sign transactions on a dedicated secure chip; ~\$50–\$200

Paper wallets: printed private key and address; immune to hacking but vulnerable to physical destruction or theft

Air-gapped computers: fully offline computers used to generate and sign; extreme security for large holdings

Key Lesson

Institutions holding large BTC positions (MicroStrategy: 214,000 BTC; BlackRock ETF: 270,000 BTC) use multi-signature cold storage with geographically distributed key shards.

A Bitcoin Paper Wallet



Source: Narayanan et al. (2016)

Bitcoin Transactions: The UTXO Model

Bitcoin Transactions: How They Actually Work

The UTXO Model

Bitcoin does not track “account balances” like a bank — it tracks **Unspent Transaction Outputs (UTXOs)**

Every transaction **consumes** previous UTXOs (inputs) and **creates** new UTXOs (outputs)

Your “balance” is the sum of all UTXOs associated with your addresses

Each UTXO can only be spent *once* (atomically consumed) — this is how double-spending is prevented

Transaction Fees

Fee = sum of inputs – sum of outputs: any “leftover” goes to the miner as fees

Fees are voluntary but necessary for timely

Example: Alice Pays Bob 1 BTC

Alice has a UTXO worth 1.5 BTC (from a prior payment)

She creates a transaction: input = 1.5 BTC UTXO; outputs = 1.0 BTC to Bob’s address + 0.4999 BTC change back to herself + 0.0001 BTC fee to miner

She signs the input with her private key

The old 1.5 BTC UTXO is destroyed; two new UTXOs (1.0 and 0.4999) are created

Key Lesson

UTXOs are like coins or banknotes — you cannot “break” a UTXO; you spend it whole and get change back. This design makes double-spend detection simple and parallelisable.

Bitcoin Transactions: Alice Pays Bob

Step 1: Bob gives Alice his public Bitcoin address (string or QR code)

Step 2: Alice opens her wallet, enters Bob's address and the amount (1 BTC)

Step 3: Alice's wallet signs the transaction with her **private key** — proving she owns the input UTXOs

Step 4: Alice broadcasts the signed transaction to the Bitcoin **peer-to-peer network**

Step 5: Miners pick up the transaction from the **mempool**, verify it, and include it in the next block

Step 6: After 6 confirmations (≈ 60 minutes), the payment is considered final and irreversible



Transaction Broadcast and Mempool

Mempool (memory pool): each full node maintains a queue of unconfirmed transactions waiting to be included in a block

Miners prioritise transactions by **fee rate** (satoshis per virtual byte); low-fee transactions may wait hours or days

Typical mempool size: 5,000–100,000 transactions; can spike to millions during congestion (e.g., Ordinals inscriptions in 2023)

Replace-by-Fee (RBF): a sender can rebroadcast a higher-fee version to bump a stuck transaction



Inside a Bitcoin Transaction

Transaction Inputs

Each input references a **previous transaction by hash** and an output index (which UTXO to spend)

Must include a **scriptSig** (unlocking script): typically a digital signature plus the public key

Multiple inputs allowed — e.g., combine three 0.3 BTC UTXOs to pay 0.9 BTC

Transaction Outputs

Each output specifies an **amount** and a **scriptPubKey** (locking script)

Most common: **P2PKH** (Pay to Public Key Hash):

`OP_DUP OP_HASH160 <address>`

`OP_EQUALVERIFY OP_CHECKSIG`

Inputs $\geq$ outputs (difference = miner fee)

Bitcoin Script

A **stack-based scripting language** embedded in every transaction; intentionally not Turing-complete (no loops) for security

Multisig (M-of-N): requires M out of N private key signatures to spend — used in escrow: buyer + seller + arbitrator, any 2-of-3 can release funds

OP_RETURN: stores up to 80 bytes of arbitrary data on the blockchain permanently (used for timestamps, NFT metadata, etc.)

Key Lesson

Bitcoin Script is Satoshi's hidden genius. It enables smart contracts far more flexible than simple payments — without the attack surface of a Turing-complete language.

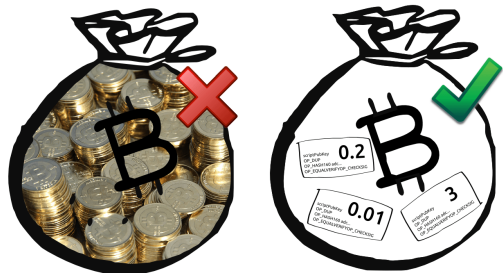
Bitcoin Balance: The UTXO Ledger

Bitcoin tracks ownership through **transaction outputs**, not account balances

Each address's balance is the **sum of all UTXOs** locked to it — verified by every full node independently

The Bitcoin UTXO set (all unspent outputs): ≈ 6 GB as of 2024 — must fit in RAM for fast validation

Privacy implication: all UTXOs and their links to prior transactions are publicly visible on the blockchain (but not linked to real identities by default)

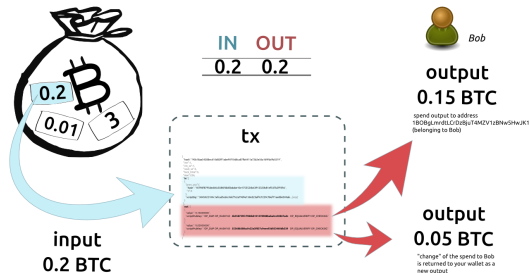


Transaction Chain: Following the Money

Every Bitcoin ever created is traceable from its **coinbase transaction** (mining reward) through every subsequent spend

Chain analysis firms (Chainalysis, Elliptic) exploit this transparency to trace stolen coins, tax evasion, and sanctions violations — over \$8B in illicit crypto identified in 2023

Users seeking privacy use **CoinJoin** (mixing multiple transactions) or privacy coins (Monero)

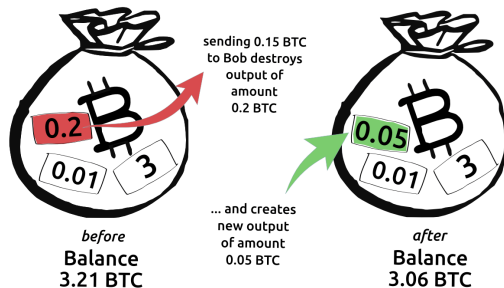


Transaction Chain: Continued

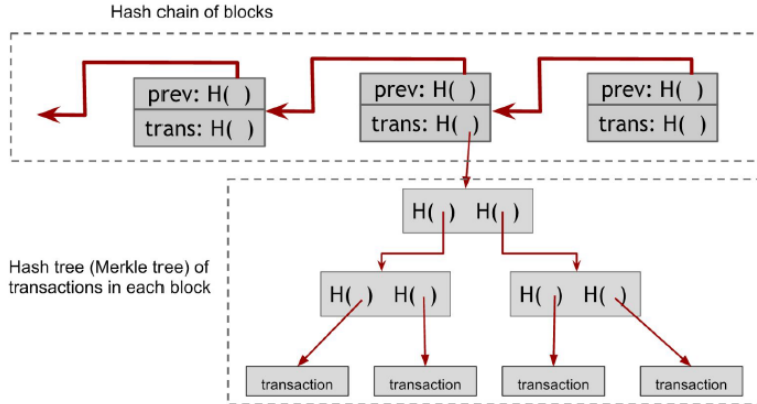
Blockchain explorers (blockchain.com, mempool.space) let anyone view any transaction in real time

Example: When MicroStrategy bought 21,000 BTC in February 2021, on-chain analysts tracked each purchase transaction in real time and identified the exchange addresses used

Pseudonymity, not anonymity: Bitcoin addresses are like pen names — one slip and the real identity is revealed



What Is Inside a Bitcoin Block?



Source: Narayanan et al. (2016)

A Real Bitcoin Block on the Chain

Block Height 503587 Blocks at depth 503587 in the bitcoin blockchain

Summary	
Height	503587 (Main chain)
Hash	000000000000000004394240d214d16bfb617e0b3dd335f324fce6148914b34
Previous Block	0000000000000000006cd0981ad2ff82c37560e2fbb2f70d53c3e0537269c2cc
Next Blocks	000000000000000000000383a7c6921fc6bc04d212d5fc08594755838b87c3cdcfb
Time	2018-01-11 00:26:59
Received Time	2018-01-11 00:26:59
Relayed By	BTCC Pool
Difficulty	1,931,136,454,487.72
Bits	402690497
Number Of Transactions	2924
Output Total	13,075.61409343 BTC
Estimated Transaction Volume	2,512.71935604 BTC
Size	1087.198 KB
Version	0x20000000
Merkle Root	9dcc12455edd288726273e32ec3bd5cb7642a19235b7e564231b0dddec991969c

Source: blockchain.info

Mining and Proof of Work

Mining: Who Maintains the Ledger?

The Role of Miners

Validate transactions: check signatures, UTXO existence, no double-spending

Order transactions: by including them in a block, miners establish a canonical ordering

Create new coins: the **coinbase transaction** creates new BTC and pays it to the miner

Collect fees: every transaction fee in the block goes to the winning miner

Mining Economics (2024)

Block reward: **3.125 BTC** $\approx$ \$200,000 at \$64,000/BTC

Plus fees: typically \$5,000–\$50,000 per block (spiked to \$2.4M during Ordinals mania)

Global mining revenue: $\approx$ \$15 billion/year at 2024 prices

The Halving Schedule

2009: **50 BTC/block** (Genesis)

2012: **25 BTC/block** (1st halving)

2016: **12.5 BTC/block** (2nd halving)

2020: **6.25 BTC/block** (3rd halving)

2024: **3.125 BTC/block** (4th halving) — April 20, 2024

2028: **1.5625 BTC/block** (5th halving, projected)

Rewards stop $\approx$ 2140; miners then supported purely by fees

Key Lesson

The halving is Bitcoin's *scheduled deflation*. Each halving has historically preceded a major price rally (2013, 2017, 2021). Whether 2024 follows suit is closely watched.

Proof of Work: The Mining Puzzle

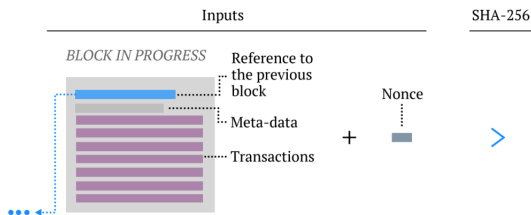
Miners must find a **nonce** (32-bit number) such that the SHA-256 hash of the block header starts with a required number of leading zeros

This is called the **target**: the hash must be numerically less than a threshold value

Expected difficulty: at Bitcoin's current difficulty, miners try $\sim 10^{23}$ hashes to find one valid block — about 100 sextillion guesses

When found: broadcast the solution; all other nodes verify with just *one* hash computation — **asymmetric effort**

Source: King, Williams and Yanofsky (2013)



Proof of Work: A Concrete Example

Find a Nonce with 3 Leading Zeros

“Hello World!**0**” → e59f8bdf... (no leading zeros)

“Hello World!**1**” → 9e2be792... (no)

“Hello World!**2**” → cacb96bf... (no)

... (keep trying)

“Hello World!**460**” → **000**7ff56bd78... (**success!**)

Difficulty Scaling

1 leading zero: ≈ 16 guesses on average

3 leading zeros: $\approx 4,096$ guesses

5 leading zeros: $\approx 1,048,576$ guesses

Bitcoin (2024): ≈ 20 leading zeros in binary (≈ 10 million trillion guesses)

Difficulty Adjustment

Every **2,016 blocks** (≈ 2 weeks), Bitcoin automatically recalibrates difficulty to target a 10-minute block time

If miners join: harder; if miners leave: easier

When China banned mining in May 2021, the global hashrate dropped 50% overnight — Bitcoin automatically adjusted downward within 2 weeks

Key Lesson

Difficulty adjustment is Bitcoin's immune system. It ensures the network stays stable regardless of how many miners participate or what hardware they use.

Code: Proof of Work Mining Simulation

```
import hashlib, time

def mine(prefix: str, difficulty: int) -> tuple:
    """Find a nonce such that SHA-256(prefix + nonce) starts with 'difficulty' zeros."""
    target = "0" * difficulty
    nonce = 0
    t0 = time.time()
    while True:
        candidate = prefix + str(nonce)
        digest = hashlib.sha256(candidate.encode()).hexdigest()
        if digest.startswith(target):
            elapsed = time.time() - t0
            return nonce, candidate, digest, elapsed
        nonce += 1

for d in [1, 2, 3, 4, 5]:
    nonce, s, h, t = mine("Hello World!", d)
    print(f"Difficulty {d}: nonce={nonce:8d}  time={t:.3f}s  hash={h[:20]}...")

# Difficulty scales exponentially: each extra zero x16 harder on average
# Bitcoin's current difficulty requires ~76 bits of leading zeros (binary)
# The entire Bitcoin network tries ~600 quintillion hashes per second (600 EH/s)
```

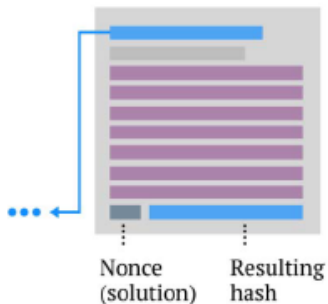
Proof of Work: Broadcast and Verification

When a miner finds a valid nonce, it broadcasts the **complete block** to the network

Every node independently verifies: just compute SHA-256 of the block header once — trivial (microseconds)

All nodes add the valid block to their copy of the chain and begin mining the **next** block on top of it

Incentive alignment: the block reward is only collectible if the block is accepted by the network — dishonest miners waste electricity for nothing



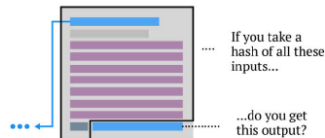
Proof of Work: Network Propagation

If two miners solve the puzzle simultaneously, a **temporary fork** occurs: different nodes see different “latest blocks”

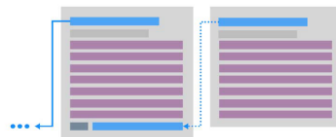
The network **resolves forks** by the longest-chain rule: whichever branch gets the next block first wins; the other branch is **orphaned**

Orphaned blocks: $\approx 5\text{--}10$ per year on Bitcoin; the miner of an orphaned block loses the reward

Transaction settlement: wait for 6 confirmations (≈ 60 minutes) for final irreversibility — Bitcoin’s de facto settlement standard



If yes, then start a new block



Hashrate: The Network's Computing Power

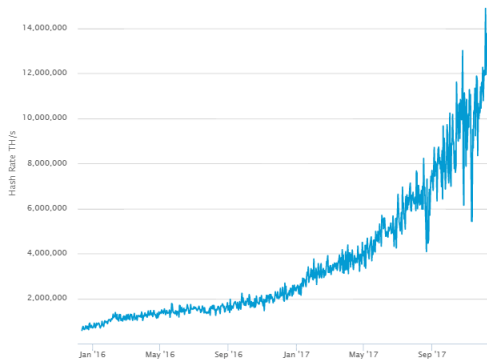
Hashrate: total SHA-256 computations per second across all miners worldwide; measured in Exahashes/second (EH/s)

Bitcoin hashrate (2024): ≈ 600 EH/s — 600 quintillion hashes per second; impossible for any single nation-state to replicate

51% attack cost: at current hashrate, an attacker would need $> \$20$ billion in ASICs and electricity to control the network for one hour

Difficulty resets every 2,016 blocks to keep block time at 10 minutes

Source: Blockchain.info



Evolution of Mining Hardware

CPU (2009–2010): any laptop could mine;
Satoshi asked Laszlo to slow GPU mining to let
CPU miners compete

GPU (2010–2012): graphics cards mine 100×
faster than CPUs; Laszlo's GPU miner changed
the game

FPGA (2011–2012): field-programmable chips,
10× more efficient than GPU

ASIC (2013–present): application-specific
chips designed solely for SHA-256; Bitmain
Antminer S19 does 95 TH/s at 3.25 kW — $10^6\times$
faster than early CPUs

Modern mining is industrial-scale: warehouses
of ASICs in Iceland, Texas, Kazakhstan



Basic Batteries

Bitmain Antminer S19 14 TH/s ASIC Bitcoin miner
+ Power supply included

[Be the first to review this item](#) | [4 answered questions](#)

List Price: \$42,000.00

Price: **\$6,399.00 & FREE Shipping**

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Only 8 left in stock - order soon.

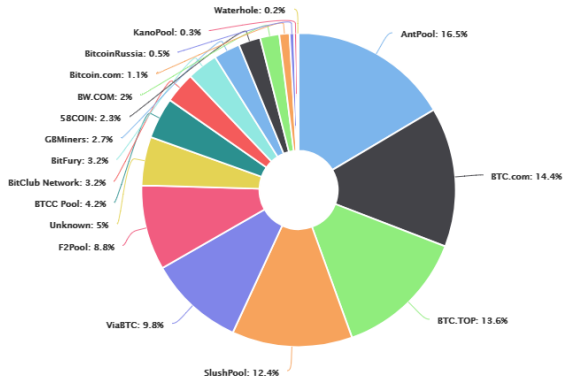
Get it as soon as Jan. 18 - 23 when you choose **Standard Shipping** at checkout.

Ships from and sold by **Antminer Direct**.

• Power Consumption 1225W (12.5TH/s batch), 1274W (13TH/s batch),
1323W (13.5TH/s batch), 1372W (14TH/s batch) (at the wall, with
Bitmain's APW3 PSU, 93% efficiency, 25°C ambient temp). Expected

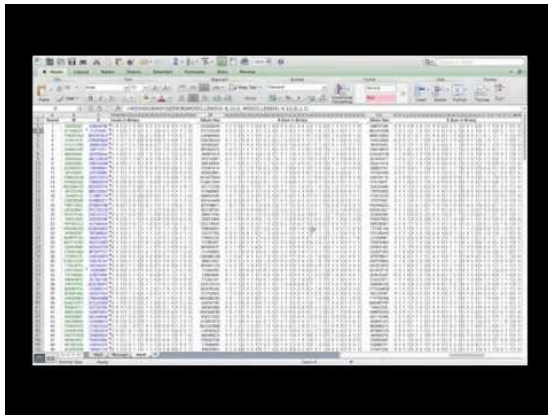
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Hashrate Distribution and Mining Pools



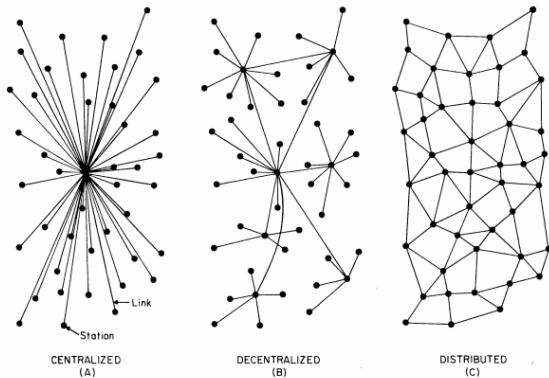
Source: blockchain.info/pools

Excel Mining of Bitcoin



Distributed Consensus and Network Security

Blockchain as a Distributed Ledger



PoW vs. PoS: Two Consensus Philosophies

Proof of Work (PoW) — Bitcoin

Miners compete to solve a SHA-256 puzzle; winner adds the next block and earns the reward

Security = economic cost of electricity and hardware; attack requires enormous physical investment

Energy: Bitcoin consumes ≈ 120 TWh/year — comparable to Argentina's entire electricity consumption

Strength: most battle-tested consensus mechanism; 15 years without a successful attack

Proof of Stake (PoS) — Ethereum (post-Merge)

Validators lock up (**stake**) a minimum of **32 ETH** as collateral; selected proportionally to stake to propose and attest blocks

The Merge (September 15, 2022): Ethereum switched from PoW to PoS, cutting energy use by **99.95%** (< 1 TWh/year)

Slashing: validators lose staked ETH for provably dishonest behaviour (e.g., signing two conflicting blocks); economic deterrent replaces electricity cost

Criticism: “rich get richer” concentration of stake

Distributed Consensus and Byzantine Generals

The Distributed Consensus Problem

Bitcoin nodes are spread worldwide with no shared clock and variable network delays

Some nodes may be **malicious** (running modified software, submitting invalid transactions)

Classic result: **Byzantine Generals Problem (Lamport, 1982)** — consensus is impossible if $\geq 1/3$ of nodes are traitors

The Byzantine Generals Problem

Generals surround a city; must all agree to attack or retreat

Can only communicate by messenger; some generals are traitors sending conflicting orders

Impossibility: cannot guarantee consensus if $\geq 1/3$ are traitors

Bitcoin's Solution

Replace messages with **proof-of-work**: a “vote” costs electricity, making Sybil attacks expensive

Probabilistic finality: after 6 blocks, reversing a transaction requires redoing all that work — economically ruinous

Invalid blocks are rejected by all honest nodes immediately (cryptographic verification)

Trustless: each node can verify everything independently without trusting any other node

Key Lesson

Nakamoto's genius: turn a provably unsolvable consensus problem into an economically solvable one by making dishonesty more expensive than honesty.

Bitcoin Consensus: Forks and the Longest Chain

How Bitcoin Achieves Consensus

Miners always work on the **longest chain** (most cumulative proof-of-work)

If two blocks are found simultaneously, a **temporary fork** forms — resolved when the next block extends one branch

6 confirmations (≈ 60 min) ensures reversal probability $< 0.1\%$ even against a 10% attacker

Double-Spending Prevention

Alice broadcasts tx1 (to Bob) and tx2 (to herself, same coins) simultaneously

Both enter the mempool; miners can only include one — they are incompatible

Only one survives in the longest chain; the other is permanently rejected

The 51% Attack

If Alice controls $>50\%$ of hashrate, she can build a secret chain faster and eventually replace the public chain

She could **reverse her own past transactions** (double-spend), but cannot steal others' coins (ECDSA is not broken)

Cost to attack Bitcoin: $> \$20$ billion — economically irrational

Smaller chains (e.g., Ethereum Classic) *have* suffered 51% attacks (2020, \$5.6M stolen)

Key Lesson

Bitcoin's security is economic, not mathematical. The honest chain wins because it is *more profitable* to follow the rules than to attack.

Protocol Upgrades: Soft Forks and Hard Forks

Soft Forks (Backward-Compatible)

New rules are *stricter*: old nodes still see new blocks as valid

If majority of hashrate upgrades, new rules are enforced network-wide

SegWit (2017): moved signature data outside the 1 MB limit; increased capacity; enabled the Lightning Network

Hard Forks (Non-Compatible)

New rules are *broadier*: old nodes reject new blocks; chain permanently splits

Both chains share history up to the fork point

Bitcoin Cash (Aug 2017): increased block size to 8 MB (later 32 MB)

Governance Challenge

Bitcoin has no CEO, no board, no vote — changes require **social consensus** among miners, developers, exchanges, and users

The block size war (2015–2017) was Bitcoin's most contentious governance crisis, resolved by the SegWit compromise

Key challenge: no built-in financial incentive to update software

Key Lesson

Bitcoin's conservatism is intentional. Changing consensus rules risks splitting the network — so changes happen slowly, with overwhelming community agreement. This is *by design*.

Bitcoin Controversies and Debates

Controversies over Bitcoin



The Conscience of a Liberal

PAUL KRUGMAN

Bitcoin Is Evil

DECEMBER 28, 2013 2:35 PM  524

What Is Bitcoin Really Worth? Don't Even Ask.

Economic View

By ROBERT J. SHILLER DEC. 15, 2017

The New York Times

How can we even start estimating the fundamental value of Bitcoin, with its astonishing market value of more than [\\$275 billion](#)? Any attempt will soon sound silly.

In short, the Bitcoin market is a marvelous case study in ambiguity and animal spirits. It is providing invaluable information about how millions of human brains process stimuli coming, in this case, from public acceptance, imagination, and innovation surrounding cryptocurrencies.



Bitcoin Controversies: Gallery I

#U.S. LEGAL NEWS SEPTEMBER 20, 2017 / 9:35 PM / 4 MONTHS AGO

JPMorgan's Dimon says bitcoin 'is a fraud'



REUTERS

NEW YORK (Reuters) - Bitcoin "is a fraud" and will blow up, Jamie Dimon, chief executive of JPMorgan Chase & Co, said on Tuesday.

Speaking at a bank investor conference in New York, Dimon said, "The currency isn't going to work. You can't have a business where people can invent a currency out of thin air and think that people who are buying it are really smart."

Dimon said that if any JPMorgan traders were trading the crypto-currency, "I would fire them in a second, for two reasons: It is against our rules and they are stupid, and both are dangerous."

Bitcoin Controversies: Gallery II

Jamie Dimon: 'I regret' calling
bitcoin a fraud



Adam Samson JANUARY 9, 2018

FINANCIAL TIMES

Bitcoin Controversies: Gallery III

Buffett on cryptocurrencies: 'I can say almost with certainty that they will come to a bad ending'

- Warren Buffett tells CNBC the recent craze over bitcoin and other cryptocurrencies won't end well.
- "When it happens or how or anything else, I don't know," the billionaire Berkshire Hathaway chief says.
- But he says he would not take a short position on bitcoin futures.

Published 8:48 AM ET Wed, 10 Jan 2018



Bitcoin: The Case Against

Intrinsic Value and Volatility

No collateral: USD is backed by the US government's taxing power; Bitcoin's value derives solely from belief and acceptance

Extreme volatility: Bitcoin fell 80% from \$68K to \$16K (Nov 2021–Nov 2022) — impossible to use as a unit of account

Speculation dominates: most BTC holders are investors, not payment users

Energy and Environment

Bitcoin mining: ≈ 120 TWh/year — comparable to Argentina's entire electricity consumption

Generates millions of tons of CO₂; ASIC hardware becomes e-waste every 2–3 years

Defenders argue mining increasingly uses stranded renewable energy

Scalability and Crime

Only 7 TPS: Bitcoin processes ≈ 7 transactions/second vs. **24,000 TPS for Visa**

Illicit use: Chainalysis estimates $\sim \$24$ B in crypto received by illicit addresses in 2023 — ransomware, darknet markets, sanctions evasion

No credit layer: no chargebacks, no credit scores, no consumer protection

Key Lesson

Bitcoin critics are not wrong on the facts. The questions are whether the benefits (censorship resistance, fixed supply, trustless settlement) outweigh these costs — and for whom.

Bitcoin: The Case For

Financial Innovation

Censorship resistance: no government or bank can block a Bitcoin transaction; used by dissidents in Venezuela, Iran, Russia to preserve savings under hyperinflation and capital controls

Inflation hedge: fixed 21M supply vs. unlimited fiat printing; Bitcoin up >10,000x since 2010 while M2 money supply doubled

Remittances: sending \$200 internationally via Bitcoin costs $\approx$ \$1 vs. $\approx$ \$12 via Western Union

24/7/365: settles on weekends, holidays, and at 3 AM — banks do not

Technology and Institutional Adoption

Blockchain innovation: the proof-of-work model spawned Ethereum, DeFi, NFTs, Layer-2 scaling, and a $\sim$ \$2T asset class

Institutional recognition: BlackRock, Fidelity, VanEck Bitcoin ETFs approved January 2024; \$12B inflows in 60 days — fastest ETF launch in US history

Corporate treasury: MicroStrategy holds 214,000 BTC; El Salvador made Bitcoin legal tender (2021)

Security record: no successful attack on Bitcoin's consensus in 15 years

Stablecoins: Bridging Crypto and Traditional Finance

What Are Stablecoins?

Definition and Purpose

Stablecoins: digital tokens designed to **maintain a stable value** relative to a reference asset (typically the US dollar)

Combine **programmability and speed** of crypto with **price stability** of fiat money

Market cap: ≈\$160 billion (2024); USDT and USDC account for >85% of exchange trading volume

Four Types of Stablecoins

Fiat-collateralised: 1:1 USD reserves (USDT, USDC)

Commodity-backed: gold or oil reserves

Crypto-collateralised: DAI (overcollateralised ETH)

Algorithmic: TerraUSD collapsed — lost \$40B in 72 hours (May 2022)

Historical Timeline

2014: **Tether (USDT)** — first large-scale USD-pegged stablecoin

2018: **USDC** launched by Circle and Coinbase; monthly audits

2020: Stablecoins become the backbone of **DeFi** (Uniswap, Aave, Compound)

2022: TerraUSD collapses, wiping \$40B

2024: PayPal USD (PYUSD) on Solana; Visa and Mastercard integrate USDC

Key Lesson

Stablecoins solved the volatility problem that prevented Bitcoin from being a practical medium of exchange. They are now the dominant settlement instrument in crypto, with daily volumes exceeding Visa.

Stablecoin Mechanisms: Maintaining the Peg

How Fiat-Backed Stablecoins Work

Collateralisation: issuer holds US Treasury bills and bank deposits equal to all tokens outstanding

Redemption: any holder can redeem 1 USDC for \$1 USD from the issuer — enforces the peg

Arbitrage: if USDC trades at \$0.99, traders buy and redeem at \$1 for profit, pushing price back to peg

Proof of reserves: Circle publishes monthly attestations audited by Deloitte; Tether faced scrutiny for mixing reserves with commercial paper

Economic Functions

Medium of exchange: transact without crypto volatility exposure

Store of value: in high-inflation countries (Argentina, Turkey), USDT surges as dollar substitute

Unit of account: DeFi protocols price collateral, loans, and yields in USDC/USDT

Cross-border settlement: send \$1M in minutes for a few dollars vs. SWIFT's 1–5 days

Key Lesson

USDT processes *more daily dollar value than PayPal* — yet it is issued by a private company with minimal regulatory oversight. This is the central tension in stablecoin policy.

Stablecoin Regulation: US and Global

United States

Regulatory landscape is fragmented: **SEC** (securities?), **CFTC** (commodities?), **OCC** (banking?), state money-transmitter licences

Proposed **GENIUS Act (2025)**: treats stablecoin issuers as insured depository institutions; mandates 1:1 reserve in T-bills or cash; requires monthly audits

Key issuers: Circle (USDC), Paxos (USDP), Tether (USDT); USDC is preferred by institutions for compliance

As of 2025: >80% of stablecoin circulation is USD-denominated

China and Asia

China **bans** private stablecoins; prioritises the **e-CNY (digital yuan)**: PBOC-issued, two-tier via commercial banks and Alipay/WeChat Pay

e-CNY: completely traceable, programmable with expiry dates, compliant with AML/KYC — the opposite of privacy-preserving crypto

Hong Kong: HKMA licensing regime for fiat-backed stablecoins; FDUSD (First Digital USD) issued in Hong Kong

EU MiCA: strict reserve management rules and issuance caps; took effect 2024

Institutional Adoption of Stablecoins

Payment Giants Integrating Stablecoins

Visa: pilot via Visa Direct allows businesses to pre-fund cross-border payouts in USDC, settling instantly vs. T+2 in traditional rails

Mastercard: expanded collaboration with Circle to enable merchants in Eastern Europe, Middle East, and Africa to settle transactions in USDC across Mastercard's network

PayPal: launched **PYUSD** on Ethereum (2023) and Solana (2024) for lower-cost transactions; backed by short-term Treasuries

Banks and Corporate Finance

JPMorgan JPM Coin: internal stablecoin for intraday dollar settlements between institutional clients; processes > \$1B/day

Tokenised deposits: banks issue blockchain-based deposit tokens for instant interbank settlement

Real-time reserves: emerging standard is on-chain proof-of-reserves (Merkle-tree verified) for instant public audit

Key Lesson

Stablecoins are no longer niche crypto instruments. They are becoming the plumbing of global dollar payments — faster, cheaper, and programmable alternatives to the correspondent banking system.

Dimension	Stablecoins	Bitcoin
Purpose	Stable value digital currency	Decentralized digital asset
Price Behavior	Low volatility	High volatility
Backing	Fiat, crypto, or algorithmic collateral	None (scarcity-based)
Issuance	Issued by firms or smart contracts	Mined via proof-of-work
Governance	Centralized or protocol-based	Fully decentralized
Supply	Flexible to maintain peg	Fixed at 21 million
Settlement	Runs on multiple blockchains	Native Bitcoin blockchain
Use Cases	Payments, DeFi, trading pairs	Store of value, transfers
Transparency	Reserve attestations or on-chain collateral	Public ledger visibility
Risks	Reserve, regulatory, or code risks	Volatility, regulation, custody risks
Regulation	Increasing oversight globally	Treated as commodity asset
Energy Use	Depends on blockchain design	High (proof-of-work)
Monetary Link	Pegged to fiat currencies	Independent of fiat money
Integration	Widely used in DeFi and payments	Limited use in mainstream payments

Review: Key Concepts in This Lecture

Cryptography Foundations

SHA-256: 256-bit one-way hash function; underpins mining, addresses, and block linking

ECDSA: secp256k1 elliptic curve signatures; private key proves ownership

Merkle tree: binary hash tree; enables $O(\log n)$ transaction verification without the full blockchain

Bitcoin Architecture

UTXO model: ownership tracked by unspent outputs; not account balances

Proof of Work: SHA-256 nonce puzzle; difficulty adjusts every 2,016 blocks

Longest chain rule: resolves forks; 6 confirmations $\approx$ final settlement

Ecosystem and Policy

Halving: block reward halves every 4 years; now 3.125 BTC; total supply capped at 21 million

The Merge: Ethereum switched from PoW to PoS (September 2022); 99.95% energy reduction

Stablecoins: fiat-backed tokens (USDT, USDC) dominate crypto trading; regulatory frameworks emerging globally

Key Lesson

Bitcoin is not just a currency — it is a proof that decentralised cryptographic consensus is possible at global scale. Every fintech application built on blockchain stands on these foundations.

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